

Looking Backward & Forward

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Today's Outline

- **Quick Segue:** Course Review
- **Review:** What we covered
- **Looking forward:** What's coming next ?
- **Project Insights**

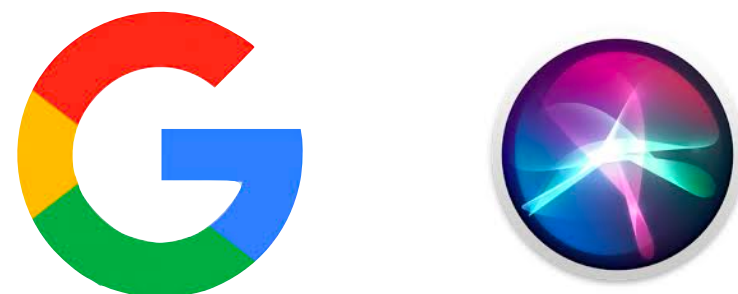
Natural Language Processing

Enabling Human-Machine Collaboration

Search Engines

Dialogue Agents

Text Generation

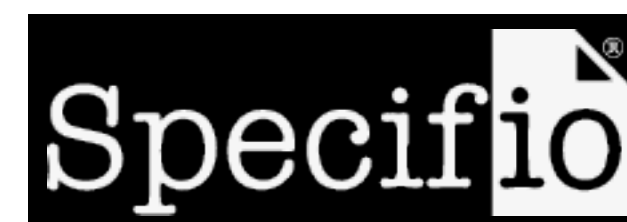


Accelerating Human-Human Communication

Machine Translation

Text Summarization

Information Extraction



Mining Human Insights

Sentiment Analysis

Motivation Analysis

Emotion Detection



Core Methods

Word Embeddings

- **Words and other tokens become vectors; no longer discrete symbols!**
 - Need to define a vocabulary of words (or token types) V that our system can assign to a vector
- Word embeddings can be learned in a self-supervised manner from large quantities of raw text
 - Learning word embeddings from scratch using labeled data for a task is data-inefficient!
- **Modern era:** embeddings learned during pretraining, and represent more granular subwords than merely words (e.g., tokenisation through BPE)

LMs & RNNs

- Language models learn to estimate the distribution over the next word given a context
- Early neural LMs (and n-gram models) suffered from **fixed context windows**
- Recurrent neural networks can **theoretically** learn to model an **unbounded context length**
 - no increase in model size because weights are shared across time steps
- Practically, however, **vanishing gradients** stop vanilla RNNs from learning useful **long-range dependencies**
- LSTMs and GRUs are variants of recurrent networks that mitigate the vanishing gradient problem
 - used for for **many sequence-to-sequence tasks**

Transformers

- **Temporal Bottleneck:** **Vanishing gradients** stop many RNN architectures from learning **long-range dependencies**
- **Parallelisation Bottleneck:** RNN states depend on previous time step hidden state, so must be **computed in series**
- **Attention:** Direct connections between output states and inputs (solves temporal bottleneck)
 - **Self-Attention:** Remove recurrence over input, allowing parallel computation for encoding
- **Transformers** use self-attention to encode sequences, but now require position embeddings to capture sequence order

Text Generation

- Text generation is the foundation of many useful NLP applications (e.g., translation, summarisation, dialogue systems)
- **Autoregressive**: models generate one token a time, using the context and previously generated tokens as inputs to generate the next token
- **Teacher forcing** is the premier algorithm for training text generators
- A variety of decoding algorithms can be used to generate text from models, each trading off **expected quality vs. diversity** in different ways.
- **Automatic evaluation** of NLG systems (content overlap, model-based, human) is difficult as most metrics fall short of reliable estimates of output quality

Modern Methods

Pretraining

- **Idea of one model base**, pretrained on large quantities of data, being applied to many tasks and applications
- **Contextual representations**: Rather than individual word embeddings, learn representations of words and sequences conditioned on the context around them
- **Many models and advantages of each**:
 - **ELMo**: Based on bidirectional LSTMs. **Good for pretrained embeddings.**
 - **GPT**: Uses a transformer decoder. **Good for generating text as a language model.**
 - **BERT**: Uses a transformer encoder. **Good for classification and sequence labelling.**
 - **BART + T5**: Pretraining sequence-to-sequence transformer models. **Extendable to all task types!**

Scale & Compression

- Scale is necessary to achieve many emergent breakthroughs: in-context learning, chain-of-thought, etc.
- Training at scale is very expensive: **months of training —> millions of \$\$\$\$**
 - Scaling laws estimate optimal model / dataset sizes for compute budget
 - Many forms of parallelism to spread computations across GPU clusters (data parallel, tensor parallel, pipe parallel)
 - **Goal:** Given the size of a cluster, use as many FLOPs as possible
- Once a large model is trained, need to optimise:
 - Memory overhead of deployment
 - Inference time
- Different model compression and inference efficiency techniques
 - Pruning, quantization, knowledge distillation, KV caching, speculative decoding

RAG & Agents AI

- Retrieval-Augmented LMs infuse knowledge from external sources into LLMs.
 - Suitable for knowledge-intensive tasks where factual accuracy is needed.
- Now, models retrieve from tools and APIs to enhance their capabilities, yielding Agentic pipelines and harnesses to execute more complex tasks
- Using external knowledge reduces how much capacity the model needs to memorize information
 - Allows for smaller models without compromising performance.

Multimodality

- Language is just “an imperfect, incomplete, and low-bandwidth serialization” of real life
 - Multimodality is the next step!
- Huge amount of multimodal data available, just waiting to be used to train a model.
- The most powerful LLMs are already multimodal (GPT, Claude, Gemini)
- Many new challenges emerge when dealing with multimodal content!

Challenges

Robustness

- Pretrained models fine-tuned on downstream tasks achieve **incredible performance** on benchmarks designed to measure language understanding
- Benchmarks are made up of datasets that are human-constructed
- Humans make (often unintentional) “mistakes” when designing datasets, allowing models to shortcut true understanding of the task in favour of easily learnable heuristics
- Designing challenging evaluations remains a primary goal natural language processing systems

Ethics

- Many ethical considerations go into designing NLP systems
 - What biases may these systems encode?
 - What groups will they exclude?
 - Will they produce toxic or misinformed content?
 - What private information can they leak about their data subjects?
 - Will their interactions with humans open up new avenues for misuse or conflict?
- Practitioners need to understand these issues to design safer systems for the benefit of all
 - Sometimes, that can mean not designing a system in the first place!

Interpretability

- Model interpretability methods allow us to explore *how* our models function when they make predictions
- **Many different approaches:** probing, local explanation, model-based explanation, and many more!
- Interpretability methods can be misleading in what they show — always be aware of the assumptions baked into your interpretation
- **Plausibility** is not the same as **faithfulness** — is this actually how your model is making a prediction? Or do you just want to believe it is?

Many challenges remain!

Legal

- **Bartz v. Anthropic:** using legally purchased copyrighted content to train LLMs constitutes fair use
 - downloading of pirated books from shadow libraries is not fair use — 1.5B judgment
- **NYT v. OpenAI:** Ongoing, but possible that NYT will win
- **GEMA v. OpenAI:** training on copyrighted song lyrics without a license is infringement
- Overall, no regulatory framework for AI — rules vary by jurisdiction.

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[CAPTION PAGE CONTINUED ON NEXT PAGE]

8 Attorneys for Defendants OPENAI, INC., a Delaware nonprofit
corporation, OPENAI, L.P., a Delaware limited partnership,
9 OPENAI GP, L.L.C., a Delaware limited liability company,
OPENAI STARTUP FUND GP I, L.L.C., a Delaware limited
10 liability company, OPENAI STARTUP FUND I, L.P., a
Delaware limited partnership, OPENAI STARTUP FUND
11 MANAGEMENT, LLC, a Delaware limited liability company

12 UNITED STATES DISTRICT COURT
13 NORTHERN DISTRICT OF CALIFORNIA
14 SAN FRANCISCO DIVISION

15 J. DOE 1 and J. DOE 2, individually and on
16 behalf of all others similarly situated,

17 Plaintiffs,

18 v.

19 GITHUB, INC., a Delaware corporation;
20 MICROSOFT CORPORATION, a Washington
corporation; OPENAI, INC., a Delaware
nonprofit corporation; OPENAI, L.P., a Delaware
21 limited partnership; OPENAI GP, L.L.C., a
Delaware limited liability company; OPENAI
22 STARTUP FUND GP I, L.L.C., a Delaware
limited liability company; OPENAI STARTUP
23 FUND I, L.P., a Delaware limited partnership;
OPENAI STARTUP FUND MANAGEMENT,
24 LLC, a Delaware limited liability company,

25 Defendants.

Case No. 4:22-cv-06823-JST
4:22-cv-07074-JST

Hon. Jon S. Tigar

CLASS ACTION

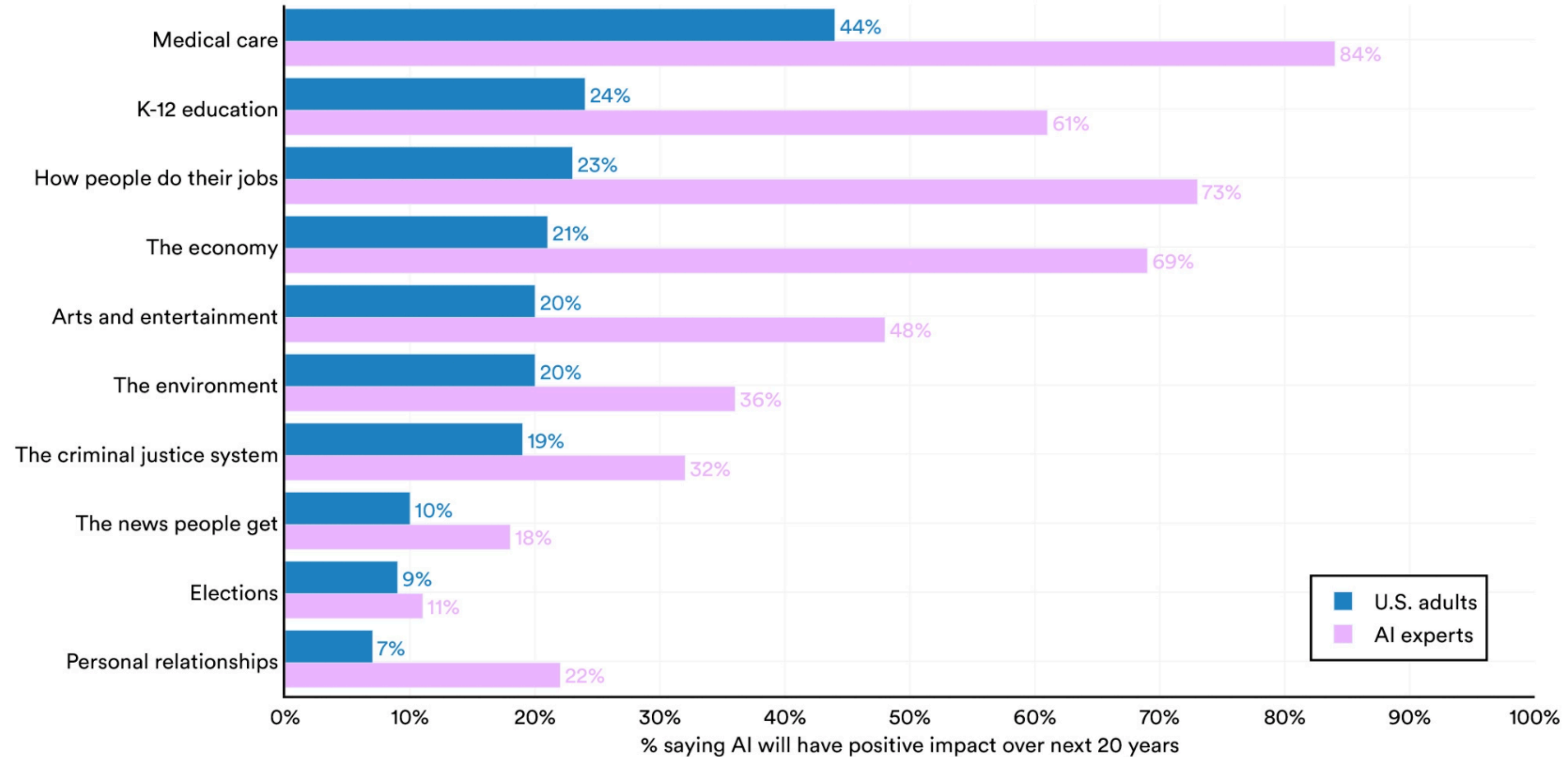
DEFENDANTS OPENAI, INC.,
OPENAI, L.P., OPENAI GP, L.L.C.,
OPENAI STARTUP FUND GP I,
L.L.C., OPENAI STARTUP FUND I,
L.P. AND OPENAI STARTUP FUND
MANAGEMENT, LLC'S NOTICE
OF MOTION AND MOTION TO
DISMISS COMPLAINT;
MEMORANDUM OF POINTS AND
AUTHORITIES

Date: May 4, 2023
Time: 2:00 p.m.
Courtroom: 6

Public Opinion

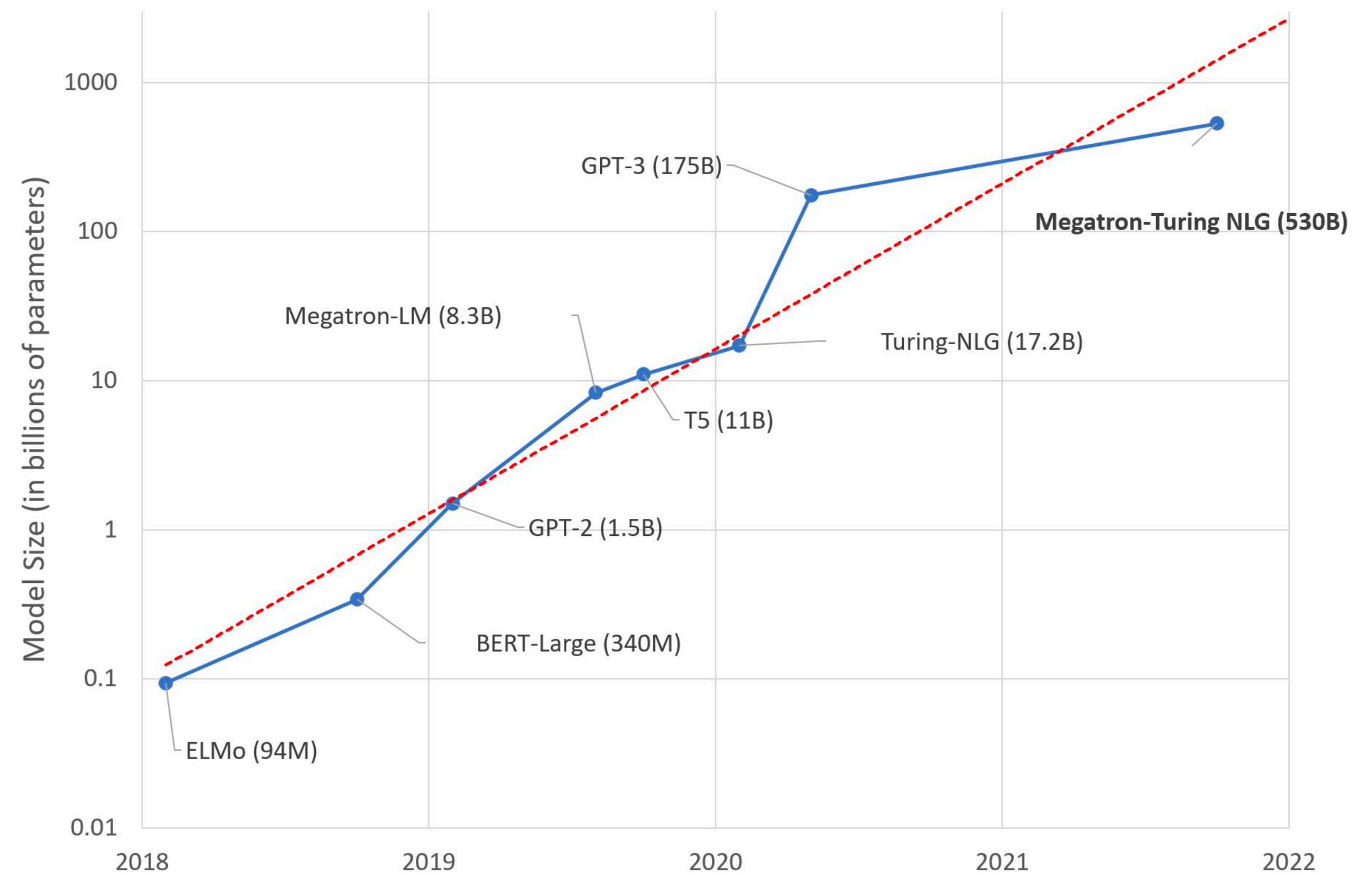
US perceptions of AI's societal impact: general public vs. experts

Source: Pew Research, 2025 | Chart: 2026 AI Index report



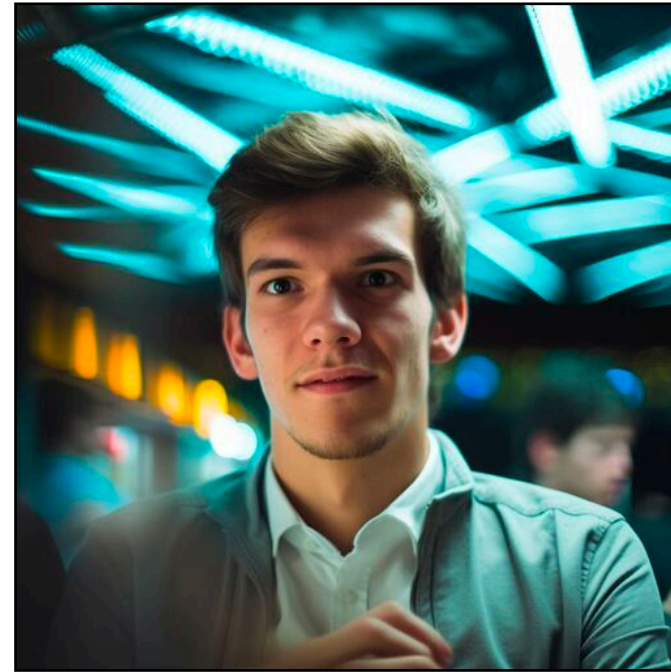
State of the field

- Advances in hardware and algorithms allowed NLP to make rapid progress
- Still lots of exciting work to be done in the field!
- **Now that the systems work**, we must be clear on how to safely and reliably deploy systems



Thanks for a great semester!

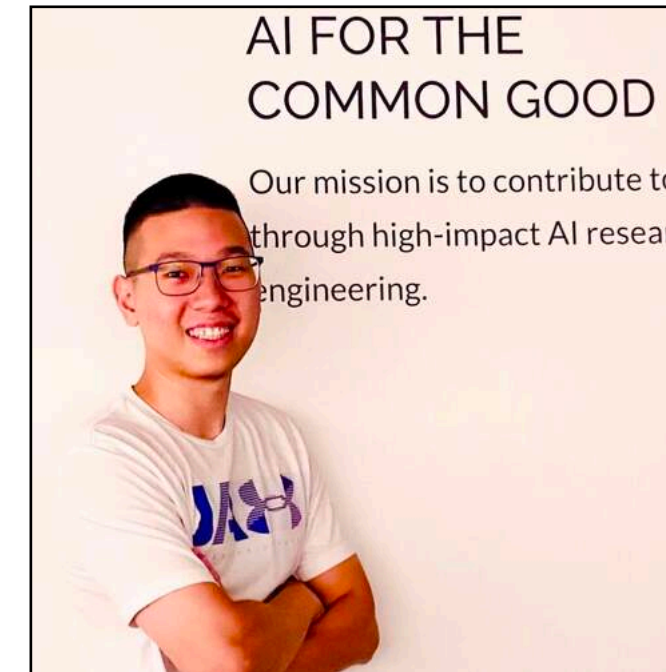
Guest Lecturers



Antonin Faure



Deniz Bayazit



Zeming Chen



Angelika Romanou



Leandro von Werra



Madhur Panwar



Sepideh Mamooler

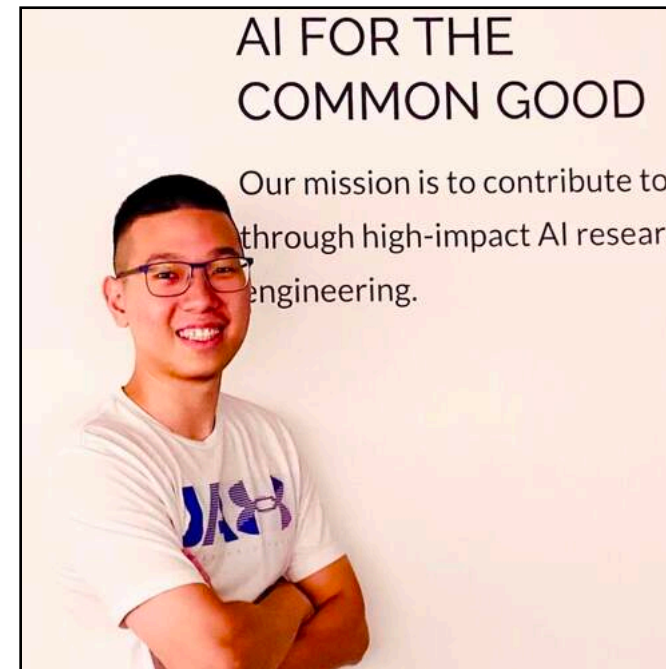


Clara Meister

Thanks for joining us!



Madhur Panwar
Head TA



Zeming (Eric) Chen
TA



Lazar Milikic
TA



Badr AlKhamissi
TA



Sepideh Mamooler
TA



Ayush Tarun
TA

Thanks for joining us!



Lysandre Costes
AE



Bich Doan
AE



Petr Grinberg
AE



Sophia Kovalenko
AE



Camille Lannoye
AE



Sebastian Maier
AE



Adrian Martínez López
AE



Quentin Sandoz
AE



Emilien Silly
AE

NLP @ EPFL

- **Natural Language Processing** Lab
 - Master's Theses, Semester Projects available every term
 - Apertus Project: Ongoing, so feel free to express interest !
- Other **NLP** courses
 - **Fall 2026**: Topics in Natural Language Processing (2 credits)
 - Paper reading, paper reviewing, discussion